

ThmSmith Stage -1 proposal:
pid_dose_response_shape / v1
Shape-closed sensitivity bounds for monotone-concave continuous dose response

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Locked parameters. This is a partial-identification anchor. The locked tuple is

$$(\text{pid_dose_response_shape}, \text{v1}, P, \theta, \mathcal{R}_{\Gamma, \text{mc}}, B_{\Gamma, \text{mc}}).$$

Here P is the observable law of (Y, A, X) , with $Y \in [0, 1]$, a continuous dose $A \in [0, 1]$, and observed covariates X . The parameter is the finite-grid dose-response vector $\theta = (\mu_0, \dots, \mu_K)$, where $0 = a_0 < \dots < a_K = 1$ are fixed design doses and $\mu_j = \mathbb{E}\{Y(a_j)\}$. The identifying restrictions $\mathcal{R}_{\Gamma, \text{mc}}$ are consistency, positivity at the design doses, the continuous-treatment marginal sensitivity model of [Jesson et al. \(2022\)](#) specialized to the grid, pointwise sharp no-shape intervals $I_j(P, \Gamma) = [L_j(P, \Gamma), U_j(P, \Gamma)]$ for each μ_j , and the shape restriction that θ is nondecreasing and discretely concave in the ordered doses. The bounding functional is

$$B_{\Gamma, \text{mc}}(P, k) = \left[\inf_{\vartheta \in \Theta_{\Gamma, \text{mc}}(P)} \vartheta_k, \sup_{\vartheta \in \Theta_{\Gamma, \text{mc}}(P)} \vartheta_k \right],$$

where

$$\Theta_{\Gamma, \text{mc}}(P) = \{\vartheta \in \mathbb{R}^{K+1} : L_j(P, \Gamma) \leq \vartheta_j \leq U_j(P, \Gamma) \forall j, \vartheta_0 \leq \dots \leq \vartheta_K, (\vartheta_{j+1} - \vartheta_j)/(a_{j+1} - a_j) \leq (\vartheta_j - \vartheta_{j-1})/(a_j - a_{j-1})\}$$

Abstract

Continuous-dose analyses often report a pointwise sensitivity band for the dose-response curve, while substantive dose-response science often assumes monotonicity and diminishing returns. This proposal asks for the sharp partial-identification envelope obtained when the continuous-treatment marginal sensitivity model of [Jesson et al. \(2022\)](#)

is intersected with monotone-concave response restrictions, without using an instrument. The routine finite-grid outer bound is the coordinate projection of a single concave sequence constrained by the pointwise sensitivity intervals. The flagship conjecture is sharper: the published continuous marginal sensitivity model admits a finite-grid normalization-pasting theorem, so every shape-compatible vector in that polytope can be generated by one full-data law rather than by dose-specific adversaries. If resolved, the result would turn modern continuous-treatment sensitivity bands into auditable minimum-effective-dose and saturation certificates under transparent shape restrictions.

1 Introduction

Why the problem is important. Dose-response decisions are rarely binary: a policy maker chooses a subsidy amount, a physician chooses a dose, and an environmental regulator chooses an exposure threshold. Recent continuous-treatment sensitivity analyses quantify how much unmeasured confounding can move a dose-response curve, but they usually deliver a family of pointwise bands. In many applications the scientific prior is also shape-based: response should be nondecreasing and have diminishing marginal returns. The methodological question is whether those two pieces of information can be combined sharply, rather than by plotting a pointwise band and then informally smoothing it.

The main finding (proposed). Theorem 1 states the routine finite-dimensional part: once the continuous marginal sensitivity model supplies sharp pointwise intervals, the monotone-concave grid envelope is the projection of an explicitly described linear-inequality polytope. Conjecture 1 is the proposed kernel: the finite-grid tilts allowed by the published continuous marginal sensitivity model can be normalized and pasted into a single hidden-confounding law whenever their implied means form a monotone-concave vector. Conjecture 2 isolates the smallest nontrivial phenomenon: with three ordered doses, a high-dose lower sensitivity endpoint can force a strictly larger lower bound at an intermediate dose through concavity, even when monotonicity alone gives no such improvement.

What is new. The closest published comparison is [Jesson et al. \(2022\)](#), which formulates continuous-valued intervention sensitivity bounds, [Marmarelis et al. \(2023\)](#), which develops partial identification for continuous

dose responses under hidden confounding, and [Baitairian et al. \(2025\)](#), which gives sharp APO bounds under a continuous sensitivity model. These papers provide pointwise continuous-dose sensitivity comparators, but they do not prove that independently sharp grid-dose adversaries can be represented by one latent treatment-density perturbation after a monotone-concave shape restriction is imposed. [Manski \(1997\)](#) and [Okumura and Usui \(2014\)](#) give sharp concave-monotone treatment-response bounds, but they do not impose a modern bounded hidden-confounding sensitivity model for continuous exposures. The novelty kernel is the compatibility claim: a pointwise sharp sensitivity band is not automatically a jointly sharp curve band, because one latent selection mechanism must rationalize all doses; the proposal asks whether the Jesson–Shalit continuous marginal sensitivity model has exactly this finite-grid pasting property.

How it positions in the literature. This is a partial-identification proposal in the Manski / shape-restriction / sensitivity-analysis cluster. It does not use an IV, a panel design, or a graphical identification algorithm. It consumes generalized-propensity and continuous-treatment sensitivity results as sources of pointwise intervals, then asks a finite convex-analytic sharpness question about the identified dose-response set under monotone-concave response.

2 Related work

[Manski \(1990\)](#) supplies the bounded-outcome partial-identification logic behind replacing unobserved potential outcomes by endpoint-compatible completions. [Manski \(1997\)](#) is the foundational shape-restricted treatment-response paper: it studies monotone, semi-monotone, and concave-monotone response functions without assuming random treatment selection. [Manski and Pepper \(2000\)](#) shows how monotone treatment selection and monotone instrumental-variable restrictions sharpen treatment-response bounds; the present proposal keeps the monotone selection intuition but replaces the instrument-style restriction with an explicit hidden-confounding sensitivity envelope. [Okumura and Usui \(2014\)](#) is the closest shape-restricted sharp-bound comparator because it combines concave-MTR and monotone treatment selection to bound mean treatment response.

[Imbens \(2000\)](#) and [Hirano and Imbens \(2004\)](#) provide the generalized-propensity foundation for dose-response estimation under observed-confounder adjustment. [Kennedy et al. \(2017\)](#) develops doubly robust nonparametric

estimation of continuous treatment effects under unconfoundedness, giving a point-identification benchmark to which the proposed bounds collapse when $\Gamma = 1$. [Westling et al. \(2020\)](#) develops causal isotonic regression for monotone dose-response curves under point identification; the present proposal asks for identified sets rather than an estimator of a point-identified monotone curve.

[Tan \(2006\)](#) introduced a marginal sensitivity model for hidden confounding that underlies much of the later bounded-confounding literature. [Yadlowsky et al. \(2022\)](#) give sharp CATE and ATE bounds under bounded unobserved confounding for binary treatment; their quantile-based extremal logic motivates the pointwise interval primitive used here. [Dorn and Guo \(2023\)](#) gives sharp sensitivity analysis for inverse-propensity weighting and shows that nonsharp sensitivity procedures can be asymptotically too wide. [Bonvini et al. \(2022\)](#) extends sensitivity analysis to marginal structural models with discrete, continuous, static, and time-varying treatments, but it targets model parameters rather than a monotone-concave curve envelope.

[Jesson et al. \(2022\)](#) proposes a continuous-treatment marginal sensitivity model and derives scalable bounds for continuous-valued interventions. [Marmarelis et al. \(2023\)](#) is the closest continuous-dose partial-identification paper, deriving hidden-confounding dose-response bounds for average and conditional average effects. [Baitairian et al. \(2025\)](#) is the closest pointwise sharpness comparator: it proves sharp bounds for the average potential outcome under a continuous sensitivity model and compares directly with Jesson et al.; the present proposal differs by asking whether several grid-dose APO bounds can be pasted through one latent law after monotone-concave shape is imposed. [Dalal and Tchetgen Tchetgen \(2025\)](#) extends Rosenbaum and marginal sensitivity ideas to endogenous continuous treatments with exposure-varying sensitivity functions and pseudo-outcome regression bounds; the present proposal is complementary because it studies shape closure and finite sharpness rather than estimation of the raw pointwise envelope.

The ThmSmith catalogue has no mature partial-ID entry for continuous-dose shape restrictions: ‘doc/API.md’ records ExactID / PartialID as a seed catalogue for future *pid** qids, and the existing documented theorem stacks are panel Q1 families. The closest in-catalogue proposals are `q1_partial.did_unobserved_trends` (aggregate-pretrend partial identification under missing cohort prehistories) and `pid_dynamic.iv_compliance.v2` (response-contribution memory bounds for a two-stage one-sided IV design). Both are partial-identification flavored, but neither concerns continuous treatment, hidden-confounding sensitivity without an instrument, or monotone-concave

dose-response shape closure.

3 Formal setup

Let P be the observable law of $O = (Y, A, X)$, where $Y \in [0, 1]$, $A \in [0, 1]$, and $X \in \mathcal{X}$. For each dose $a \in [0, 1]$, let $Y(a)$ denote the potential outcome and let $\mu(a) = \mathbb{E}\{Y(a)\}$. Fix a finite grid $0 = a_0 < \dots < a_K = 1$ and write $\mu_j = \mu(a_j)$, $\theta = (\mu_0, \dots, \mu_K)$, and $h_j = a_j - a_{j-1}$.

The bounded-confounding primitive is the continuous-treatment marginal sensitivity model of [Jesson et al. \(2022\)](#), specialized to the finite grid. Let $e(a | x) = f_{A|X}(a | x)$ denote the observed generalized propensity density, and for a full-data law let $e_j^\dagger(a_j | x, y) = f_{A|X, Y(a_j)}(a_j | x, y)$ denote the complete propensity density at the same grid dose after conditioning on the potential outcome $Y(a_j) = y$. We use the complete-over-nominal ratio convention

$$R_j(x, y) = \frac{e_j^\dagger(a_j | x, y)}{e(a_j | x)}, \quad \Gamma^{-1} \leq R_j(x, y) \leq \Gamma,$$

with both densities evaluated at $A = a_j$ and with the conditioning object $Y(a_j)$, not $Y(a_\ell)$ for $\ell \neq j$. The inverse observed tilt is

$$w_j(y, x) = R_j(x, y)^{-1} = \frac{e(a_j | x)}{e_j^\dagger(a_j | x, y)}.$$

Write $\mathcal{W}_{\Gamma, j}(P)$ for the set of measurable tilting functions $w_j(Y, X)$ over the regular conditional law of (Y, X) given $A = a_j$ such that $\Gamma^{-1} \leq w_j \leq \Gamma$ and $\mathbb{E}\{w_j(Y, X) | A = a_j, X = x\} = 1$ for P_X -almost every x . The no-shape pointwise interval is

$$I_j(P, \Gamma) = [L_j(P, \Gamma), U_j(P, \Gamma)]$$

where

$$L_j(P, \Gamma) = \inf_{w_j \in \mathcal{W}_{\Gamma, j}(P)} \int \mathbb{E}\{Y w_j(Y, X) | A = a_j, X = x\} dP_X(x)$$

and $U_j(P, \Gamma)$ is the corresponding supremum. These endpoints are assumed sharp before imposing shape across doses. The nontrivial open issue is whether a tuple $(w_0, \dots, w_K)$ of pointwise tilts whose means satisfy the shape inequalities can be pasted into one full-data law with a single latent

U and one conditional treatment density $f(a \mid X, U)$ such that, for every grid dose, the induced complete-over-nominal ratios $f(a_j \mid X, U)/e(a_j \mid X)$ obey the same Γ -bound and induce the displayed $Y(a_j)$ -conditioned ratio after marginalizing U .

The shape-restricted identified set on the grid is

$$\Theta_{\Gamma, \text{mc}}(P) = \left\{ \vartheta \in \mathbb{R}^{K+1} : \begin{array}{ll} L_j(P, \Gamma) \leq \vartheta_j \leq U_j(P, \Gamma) & \forall j, \\ \vartheta_{j-1} \leq \vartheta_j & \forall 1 \leq j \leq K, \\ (\vartheta_j - \vartheta_{j-1})/h_j \geq (\vartheta_{j+1} - \vartheta_j)/h_{j+1} & \forall 1 \leq j < K \end{array} \right\}.$$

For a target grid index k , the proposed bounding functional is the coordinate projection

$$B_{\Gamma, \text{mc}}(P, k) = [\underline{\mu}_k^{\text{mc}}(P, \Gamma), \overline{\mu}_k^{\text{mc}}(P, \Gamma)] = \left[\inf_{\vartheta \in \Theta_{\Gamma, \text{mc}}(P)} \vartheta_k, \sup_{\vartheta \in \Theta_{\Gamma, \text{mc}}(P)} \vartheta_k \right].$$

locked tuple:

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(pid_dose_response_shape, v1,
 P, theta, R_{Gamma,mc}, B_{Gamma,mc})
P: observable law of (Y,A,X), with Y in [0,1] and A in [0,1].
theta: finite-grid dose-response vector (mu_0,...,mu_K),
      mu_j = E[Y(a_j)] at fixed doses 0=a_0<...<a_K=1.
R_{Gamma,mc}: consistency, positivity at grid doses, the
      Jesson-Shalit continuous marginal sensitivity model specialized
      to the grid, sharp pointwise no-shape intervals
      I_j(P,Gamma)=[L_j(P,Gamma),U_j(P,Gamma)], and nondecreasing
      discrete concavity of theta in the ordered doses.
B_{Gamma,mc}: coordinatewise sharp lower and upper projections of
      Theta_{Gamma,mc}(P) onto the requested dose index k.
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4 Assumptions

Assumption 1 (Observed support and bounded outcomes). The observable law P is a probability law for (Y, A, X) with $Y \in [0, 1]$, $A \in [0, 1]$, and fixed grid $0 = a_0 < \dots < a_K = 1$ with $K \geq 2$.

Assumption 2 (Consistency). If $A = a_j$, then the observed outcome equals the corresponding potential outcome at that dose, $Y = Y(a_j)$, for each grid dose a_j .

Assumption 3 (Grid positivity). The observed generalized propensity at every grid dose is positive on the relevant covariate support, so the selected continuous-treatment sensitivity model defines finite pointwise bounds $L_j(P, \Gamma)$ and $U_j(P, \Gamma)$.

Assumption 4 (Continuous marginal sensitivity model). For a fixed $\Gamma \geq 1$, the hidden-confounding restriction is the continuous-treatment marginal sensitivity model of [Jesson et al. \(2022\)](#) on the grid $a_0, \dots, a_K$, using the ratio orientation in Section 3. At dose a_j , the complete-over-nominal propensity-density ratio is $R_j(X, Y(a_j)) = e_j^\dagger(a_j \mid X, Y(a_j))/e(a_j \mid X)$ and must lie in $[\Gamma^{-1}, \Gamma]$. The induced inverse tilt $w_j = R_j^{-1}$ belongs to $\mathcal{W}_{\Gamma,j}(P)$, is normalized conditionally on X among units with $A = a_j$, and computes the APO at a_j by integrating over P_X . A joint full-data law in the model must use one latent variable U and one conditional treatment density $f(a \mid X, U)$ whose grid-dose complete-over-nominal ratios all satisfy the common Γ -bound.

Assumption 5 (Pointwise sharp bounded-confounding envelope). For a fixed $\Gamma \geq 1$, Assumption 4 implies, before imposing shape across doses, the sharp pointwise statement

$$\mu_j \in I_j(P, \Gamma) = [L_j(P, \Gamma), U_j(P, \Gamma)]$$

for each grid dose a_j , and every value in $I_j(P, \Gamma)$ is attainable by some full-data law satisfying the same pointwise sensitivity restriction at dose a_j .

Assumption 6 (Monotone-concave dose response). The grid dose-response vector $\theta = (\mu_0, \dots, \mu_K)$ is nondecreasing and discretely concave:

$$\mu_{j-1} \leq \mu_j \quad (1 \leq j \leq K), \quad \frac{\mu_j - \mu_{j-1}}{a_j - a_{j-1}} \geq \frac{\mu_{j+1} - \mu_j}{a_{j+1} - a_j} \quad (1 \leq j < K).$$

Assumption 7 (Nonempty shape-compatible envelope). The polytope $\Theta_{\Gamma, \text{mc}}(P)$ defined in Section 3 is nonempty.

5 Main results

Theorem 1 (Theorem 1: finite-grid shape-closed outer bound). *Under Assumptions 1–7, every feasible dose-response vector under the continuous marginal sensitivity pointwise intervals and monotone-concave shape restriction belongs to $\Theta_{\Gamma, \text{mc}}(P)$. Hence, for each target dose index k ,*

$$\mu_k \in \left[\inf_{\vartheta \in \Theta_{\Gamma, \text{mc}}(P)} \vartheta_k, \sup_{\vartheta \in \Theta_{\Gamma, \text{mc}}(P)} \vartheta_k \right],$$

and the two endpoints are attained by vertices of the finite linear-inequality polytope $\Theta_{\Gamma, \text{mc}}(P)$.

Why this is new and worth studying. The closest published papers split the ingredients: [Marmarelis et al. \(2023\)](#), [Jesson et al. \(2022\)](#), and [Dalal and Tchetgen Tchetgen \(2025\)](#) give continuous-treatment sensitivity bands, while [Manski \(1997\)](#) and [Okumura and Usui \(2014\)](#) give shape-restricted treatment-response bounds. This theorem is not the field-level novelty claim; it records the named finite LP certificate that the pasting conjecture must make sharp. Its value is diagnostic: it tells a reviewer exactly which coordinate projection would be solved, where shape restrictions tighten a reported sensitivity band, and where nonemptiness fails.

Conjecture 1 (Conjecture 1: finite-grid CMSM normalization-pasting sharpness (NOT YET PROVED)). *Under Assumptions 1–7, the continuous marginal sensitivity model in Assumption 4 has the following normalization-pasting property. For every $\vartheta \in \Theta_{\Gamma, \text{mc}}(P)$, there exist grid-dose tilts $w_j \in \mathcal{W}_{\Gamma, j}(P)$ with*

$$\int \mathbb{E}\{Y w_j(Y, X) \mid A = a_j, X = x\} dP_X(x) = \vartheta_j$$

that are induced by one full-data law with a single latent variable U , one conditional treatment density $f(a \mid X, U)$, the observed law P , and the common complete-over-nominal Γ -bound at all grid doses. Consequently, $\Theta_{\Gamma, \text{mc}}(P)$ is the sharp finite-grid identified set for θ , and $B_{\Gamma, \text{mc}}(P, k)$ is the sharp partial-identification interval for each μ_k .

Why this is new and worth studying. This is the exact gap left by the closest continuous-dose sensitivity papers: pointwise sharpness in [Jesson et al. \(2022\)](#), [Marmarelis et al. \(2023\)](#), and [Baitairian et al. \(2025\)](#) does not automatically imply joint curve sharpness, because a single latent confounding law must rationalize all grid doses at once. Baitairian et al. are especially close on the published axis because their APO bounds are sharp under a continuous sensitivity model, but their statement is pointwise in the dose and does not add a monotone-concave finite-grid pasting theorem. If the conjecture is true, existing continuous-treatment sensitivity methods can be upgraded into sharp shape-restricted curve bounds by solving only a finite convex projection; if it is false, the counterexample would identify a previously hidden cross-dose compatibility restriction in the published continuous marginal sensitivity model.

Conjecture 2 (Conjecture 2: three-dose concavity phase transition (NOT YET PROVED)). *Suppose $K = 2$, $a_0 < a_1 < a_2$, Assumptions 1–7 hold, and Conjecture 1 holds for the same P and Γ . There are observable laws P and sensitivity levels $\Gamma > 1$, with $L_0(P, \Gamma) \leq L_1(P, \Gamma)$, such that monotonicity alone leaves the lower bound at a_1 equal to $L_1(P, \Gamma)$, while monotone concavity raises it to*

$$\underline{\mu}_1^{\text{mc}}(P, \Gamma) = \max \left\{ L_1(P, \Gamma), \frac{a_2 - a_1}{a_2 - a_0} L_0(P, \Gamma) + \frac{a_1 - a_0}{a_2 - a_0} L_2(P, \Gamma) \right\} > L_1(P, \Gamma).$$

The strict inequality is feasible whenever the chord through the endpoint lower sensitivity bounds lies above the pointwise middle-dose lower bound and below the corresponding upper feasibility region.

Why this is new and worth studying. Westling et al. (2020) shows how monotonicity can stabilize estimation when the curve is point identified, and Manski (1997) shows that concavity can sharpen response bounds without modeling selection. This conjecture isolates a smaller and testable phenomenon for hidden-confounding sensitivity: a high-dose lower guarantee can propagate backward to an intermediate dose through diminishing returns. That gives practitioners a finite certificate for minimum-effective-dose claims that cannot be read from pointwise sensitivity bands.

Theorem 2 (Theorem 2: no-hidden-confounding collapse). *Under Assumptions 1–5, if $\Gamma = 1$ and the continuous marginal sensitivity model reduces to observed-confounder identification at every grid dose, so $L_j(P, 1) = U_j(P, 1) = \mu_j^0(P)$, then $\Theta_{1,\text{mc}}(P) = \{\mu^0(P)\}$ when the identified dose-response vector is monotone-concave and $\Theta_{1,\text{mc}}(P) = \emptyset$ when that vector violates the imposed shape restriction. In the nonempty branch of Assumption 7, only the singleton case can occur.*

Why this is new and worth studying. The closest point-identification benchmark is Imbens (2000) together with the continuous-treatment estimation literature such as Kennedy et al. (2017). This theorem is a sanity theorem rather than the novelty kernel: it verifies that the proposed partial-ID object collapses to the usual generalized-propensity dose-response curve when the hidden-confounding radius is zero and that shape is treated as a falsifiable restriction, not as a smoothing convention.

6 Sanity-check corollaries

Corollary 1 (Reduction to coordinatewise interval-plus-concavity projection). *Under Assumptions 1, 6, and 7, if the sensitivity model is used only through externally supplied coordinate intervals $I_j = [L_j, U_j]$, then $B_{\Gamma, \text{mc}}$ is exactly the finite-dimensional projection of the interval box $\prod_{j=0}^K [L_j, U_j]$ intersected with the monotone-concave cone. This is the grid-level convex-analytic analogue of the shape-restricted envelope studied by [Manski \(1997\)](#), without claiming to encode Manski’s individual realized-dose response-function information.*

Under the named assumptions, the statement collapses to interval projection because the only remaining restrictions on θ are membership in the coordinate box and the two linear shape systems.

Example 1 (Three-dose tightening certificate). Let $K = 2$, $a_0 = 0$, $a_1 = 1/2$, $a_2 = 1$, and take pointwise lower endpoints $L_0 = 0.20$, $L_1 = 0.20$, $L_2 = 0.80$ with upper endpoints $U_0 = 0.40$, $U_1 = 0.90$, and $U_2 = 0.95$. Monotonicity alone permits $\mu_1 = 0.20$, but monotone concavity forces $\mu_1 \geq (\mu_0 + \mu_2)/2 \geq 0.50$, and the vector $(0.20, 0.50, 0.80)$ is feasible.

Under Assumptions 1, 6, and 7, the example collapses to one line of algebra: for equally spaced doses, discrete concavity is $2\mu_1 \geq \mu_0 + \mu_2$, so endpoint lower bounds L_0 and L_2 impose the chord lower bound at the middle dose.

Corollary 2 (Reduction to the unconfounded dose-response curve). *Under Assumptions 1–7, if $\Gamma = 1$, $L_j(P, 1) = U_j(P, 1) = \mu_j^0(P)$, and $\mu^0(P)$ is monotone-concave, then $B_{1, \text{mc}}(P, k) = \{\mu_k^0(P)\}$ for every grid index k .*

Under the named assumptions, the statement collapses to the generalized-propensity point-identification result because each coordinate interval is a singleton and the shape inequalities are already satisfied.

7 Proof-sketch route

The mathematical route is finite convex analysis plus one named sensitivity-model realization theorem. Theorem 1 follows by writing the pointwise interval, monotonicity, and discrete-concavity restrictions as finitely many linear inequalities and applying the elementary fact that a nonempty compact polytope has coordinate extrema at vertices. For Conjecture 1, fix the

complete-over-nominal CMSM ratio $R_j = e_j^\dagger/e$ from Section 3; the required theorem is a finite-grid normalization-pasting result showing that any tuple of conditionally normalized inverse tilts $w_j = R_j^{-1} \in \mathcal{W}_{\Gamma,j}(P)$ with monotone-concave APO means can be represented by one latent U and one conditional density $f(a \mid X, U)$, while preserving the observed conditional treatment density after integrating out U . The $K = 2$ construction should first split each conditional law $(Y, X) \mid A = a_j$ into at most two latent strata that attain the desired inverse-tilt mean, then check whether the three dose-specific strata can be coupled through a common latent partition without changing P_X or the observed generalized propensity. The boundary algebra is a three-dose calculation: for equal spacing, $2\mu_1 \geq \mu_0 + \mu_2$, so the middle lower endpoint is the maximum of the monotone lower endpoint and the endpoint chord; unequal spacing replaces the arithmetic mean by the affine chord in Conjecture 2. The cases to enumerate explicitly are empty intersections, upper endpoints that block the chord witness, and whether the tilting functions that attain the three coordinate means can be chosen from a common latent partition.

8 Honest scope flags

The finite-polytope outer-bound statement is routine algebra once the pointwise sensitivity intervals are accepted as inputs. The joint-sharpness statement is deliberately labelled as a conjecture because pointwise sharp sensitivity bounds may fail to paste across doses under the continuous marginal sensitivity model. The three-dose phase-transition formula is mathematically plausible at the interval-polytope level, but its full causal sharpness depends on Conjecture 1. This v3 draft fixes the sensitivity ratio orientation to the complete-over-nominal CMSM convention and adds [Baitairian et al. \(2025\)](#) as the closest pointwise-sharp continuous sensitivity comparator. The dose-varying formulation of [Dalal and Tchetgen Tchetgen \(2025\)](#) and the marginal structural model sensitivity class of [Bonvini et al. \(2022\)](#) are related work rather than theorem substrate. The v1 Manski sanity corollary has been withdrawn and replaced by the safer coordinatewise interval-plus-concavity reduction.

9 Iteration trail

- v1 (cold-start) — initial draft of a partial-ID proposal for monotone-concave shape closure of continuous-treatment sensitivity intervals; prior

verdict: none.

- v2 (revise) — instantiated the Jesson–Shalit continuous marginal sensitivity model, made normalization-pasting the conjecture rather than an assumption, repaired the $\Gamma = 1$ empty-set branch, and replaced the Manski sanity reduction; prior verdict: REVISE.
- v3 (revise) — pinned the CMSM density-ratio orientation and conditioning objects, added Baitairian et al. as the pointwise-sharp continuous-sensitivity comparator, and sketched the $K = 2$ latent-partition pasting route; prior verdict: REVISE.

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